

Chapter-5 Implementation of Discrete-time System

LTI System can be classified according to the length of the impulse response. Thus, discrete-time LTI Systems can be Finite-impulse Response and infinite impulse Response. If the impulse response is of finite duration, then the system is known as finite duration impulse response (FIR) System. It is described by using difference equation.

If the impulse response is of infinite duration then the system is called Infinite duration impulse response (IIR) System.

FIR System	IIR System.
1. Finite duration impulse	1. Infinite duration impulse response
2. For causal FIR System, $h(n)=0$ for $n < 0$ & $n \geq M-1$ then, Convolution output becomes $y(n) = \sum_{k=0}^{M-1} h(k) x(n-k)$	2. For IIR System, $h(n)=0$ for $n < 0$, then Convolution output becomes $y(n) = \sum_{k=0}^{\infty} h(k) x(n-k)$
3. FIR system has finite memory length.	3. IIR System has infinite memory length.
4. FIR system can be realized with finite no. of addition, multiplication & memory location	4. IIR System can't be realized with Convolution sum, because it requires infinite no. of add ⁿ & mulip ⁿ . So, described by difference eq ⁿ .
5. Moving Average System $y(n) = \frac{1}{M} \sum_{k=0}^M x(n-k)$	5. Cumulative Average System $y(n) = \frac{1}{n+1} \sum_{k=0}^n x(k)$
6.	

Structure of FIR System;

FIR system is described by the difference equation

$$y(n) = \sum_{k=0}^{M-1} b_k x(n-k)$$

And the system function is

$$H(z) = \sum_{k=0}^{M-1} b_k z^{-k}$$

where M is length of FIR system.

i) Direct form structure (canonical form)

for a causal FIR system

$$y(n) = \sum_{k=0}^{M-1} h(k) x(n-k]$$

$$\therefore y(n) = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) + \dots$$

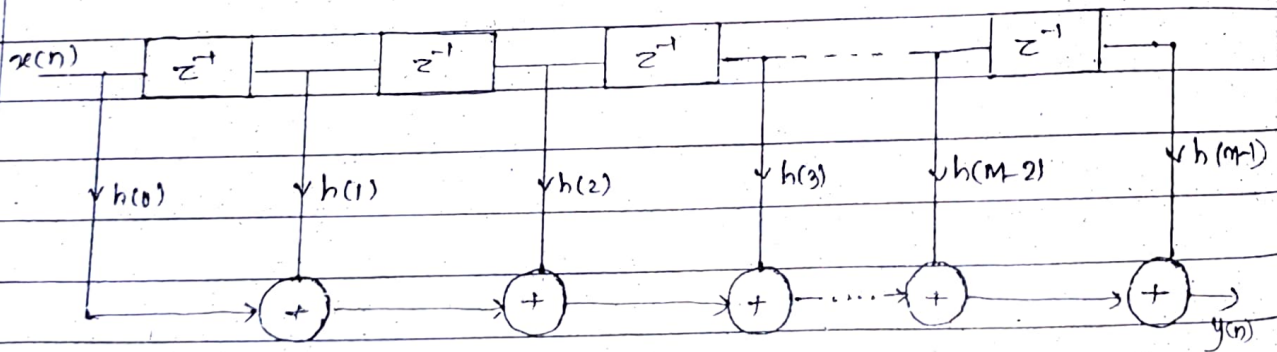


Fig: Direct form realization of FIR system.

ii) Cascade - Form structures.

Here,

$$H(z) = \prod_{k=1}^K H_k(z)$$

where, $H_k(z) = b_{k0} + b_{k1} z^{-1} + b_{k2} z^{-2}, \quad k=1, 2, \dots, K.$

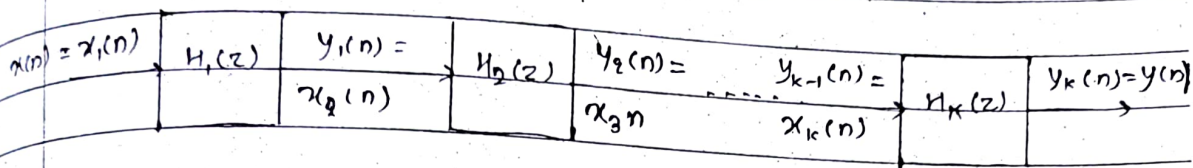


Fig. cascade realization of FIR system.

Structure of IIR System

The causal IIR system are characterized by constant coeff. diff eqn i.e

$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

or equivalent

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

From these eqns we observed that realization of IIR system involves a recursive computational algorithm.

- i) Direct forms I and II structure
- ii) cascade realization structure
- iii) Parallel realization structure

i) Direct form I & II structures:

we have

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$$

$H(z)$ can be viewed as two systems in cascade.

$$H_1(z) = \sum_{k=0}^M b_k z^{-k}$$

$$\& H_2(z) = \frac{1}{1 + \sum_{k=1}^N a_k z^{-k}}$$

By attaching all-pole system in cascade with $H_1(z)$, we obtain direct form - I shown in fig. below.

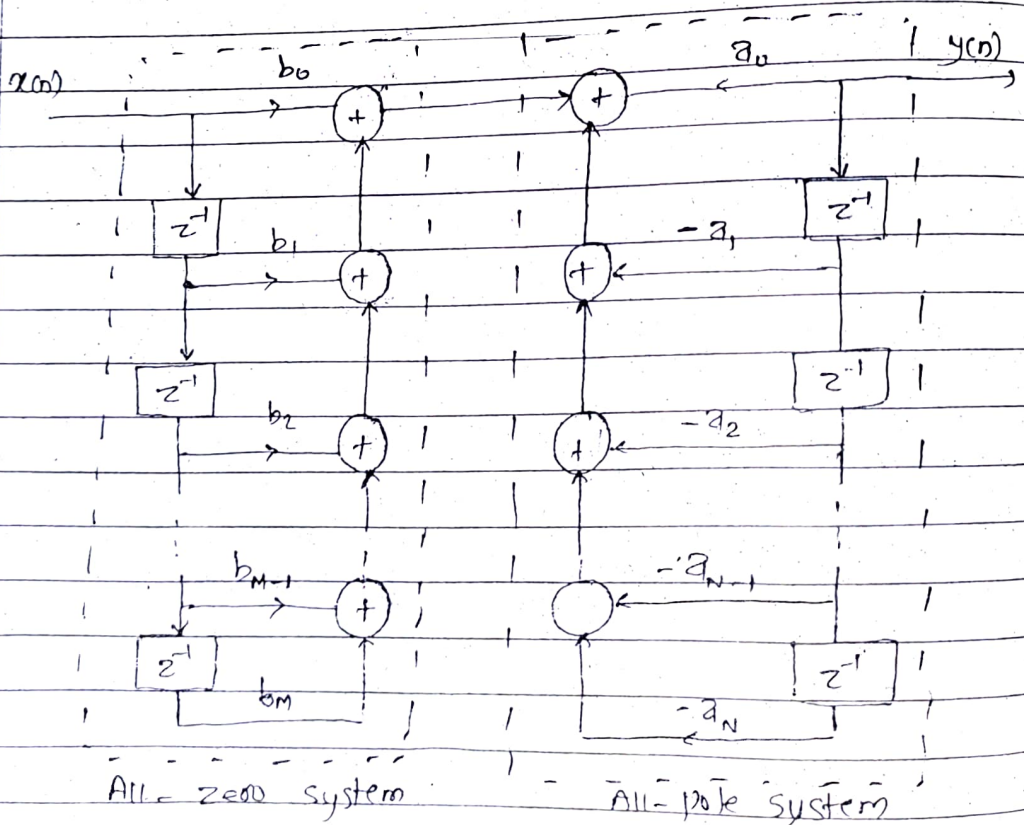


fig. Direct form I - structure.

differe eqⁿ for all pole filter (system) is

$$w(n) = - \sum_{k=1}^N a_k w(n-k) + x(n) \quad \text{--- (i)}$$

since $w(n)$ is the input to all zero system, its output is

$$y(n) = \sum_{k=0}^M b_k w(n-k) \quad \text{--- (ii)}$$

which (eqⁿ (i) & (ii)) shown in fig. below.

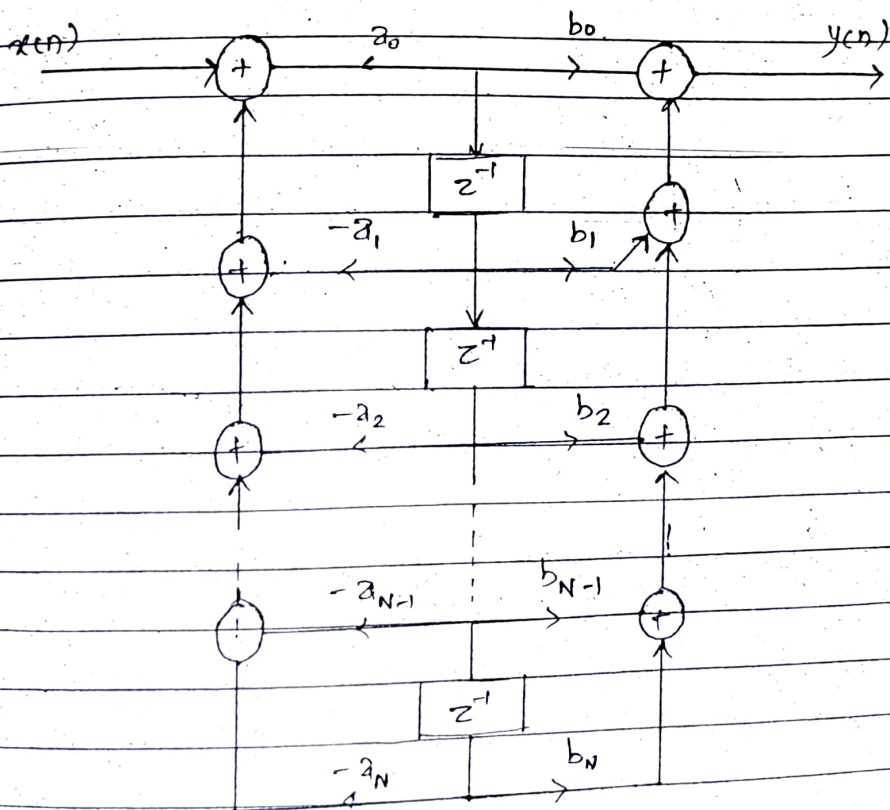


Fig. Direct form II realization ($N=M$)

ii) Cascade structure:

Let

$$H(z) = \prod_{k=1}^K H_k(z) = H_1(z) \cdot H_2(z) \cdot H_3(z) \cdots$$

where, K is the integer part $\frac{N+1}{2}$.

$H(z)$ has the general form

$$H_k(z) = \frac{b_{k0} + b_{k1}z^{-1} + b_{k2}z^{-2}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}}$$

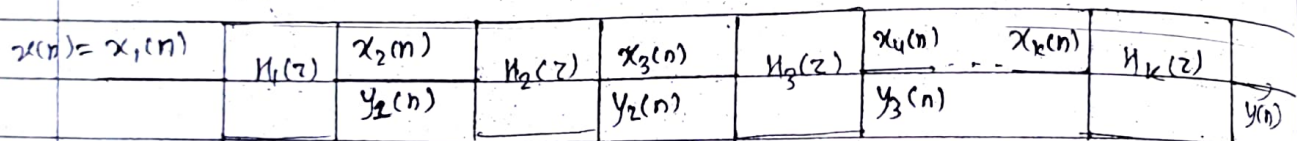


Fig: cascade structure

iii) parallel structure:

$$H(z) = C + \sum_{k=1}^N \frac{A_k}{1 - p_k z^{-1}}$$

$$\text{Also, } H_k(z) = \frac{b_{k0} + b_{k1}z^{-1}}{1 + a_{k1}z^{-1} + a_{k2}z^{-2}}$$

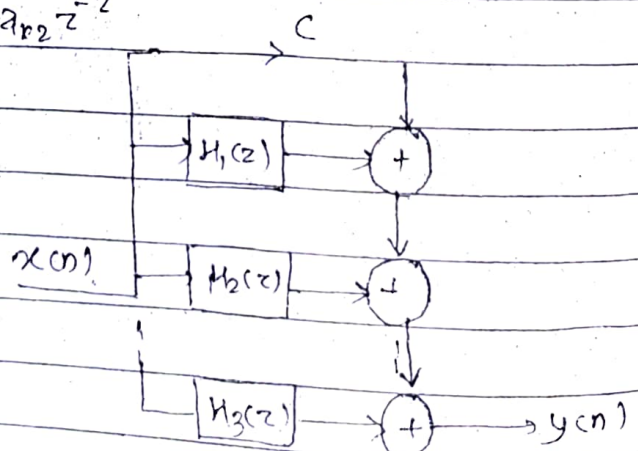


Fig: parallel structure

Lattice structure for FIR system:

Lattice filters are used extensively in digital speech processing and in the implementation of adaptive filters.

Let us begin the development by considering a sequence of FIR filters with system functions.

$$H_m(z) = A_m(z), \quad m=0,1,2,\dots,M-1$$

where,

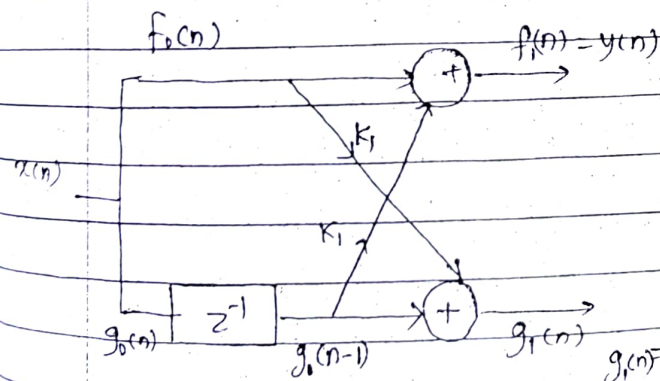
$$A_m(z) = 1 + \sum_{k=1}^m a_m(k) z^{-k} \quad \text{is the polynomial by definition.}$$

where,

$$A_0(z) \equiv 1, \quad a_m(k) \rightarrow \text{filter coefficients.}$$

If $x(n)$ is the input sequence to the filter $A_m(z)$ and $y(n)$ is the output of sequence, we have

$$y(n) = x(n) + \sum_{k=1}^m a_m(k) \cdot x(n-k)$$



where, (From the figure)

$$f_0(n) = g_0(n) = x(n)$$

$$f_1(n) = f_0(n) + k_1 g_0(n-1) = x(n) + k_1 x(n-1)$$

$$g_1(n) = g_0(n) + k_1 f_0(n) = x(n) + k_1 x(n-1)$$

Fig: Single-stage lattice filter.

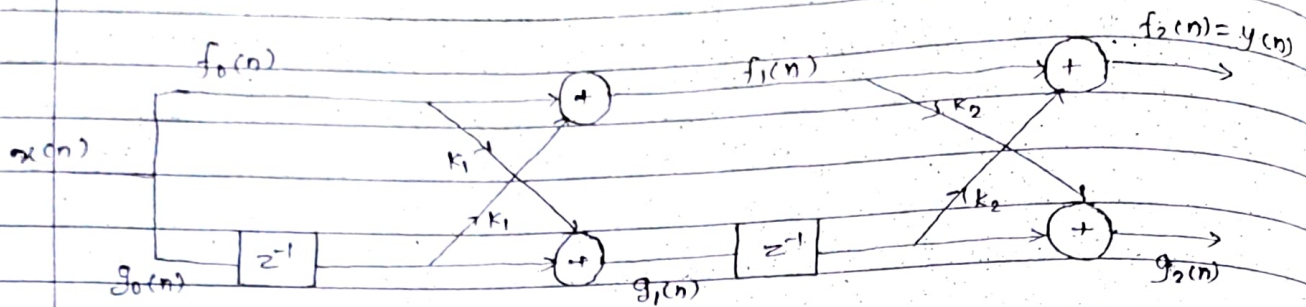
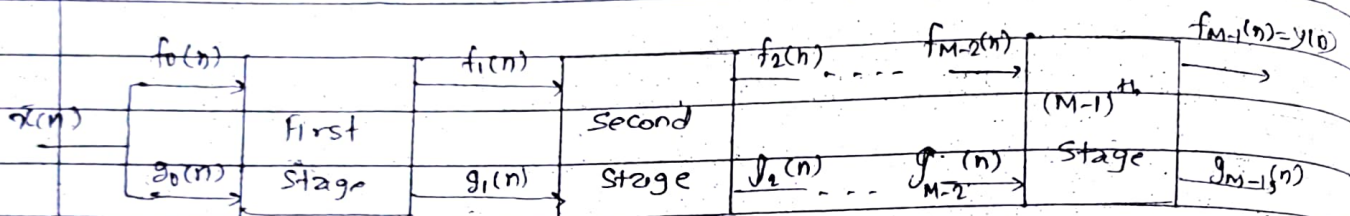


Fig: Two stage lattice filter



Conversion of lattice coeff. to direct form filter coeff.

The direct form FIR filter coeff. $a_m(k)$ can be obtained from lattice coeff K_i by using the following relations.

$$A_0(z) = B_0(z) = 1$$

$$A_m(z) = A_{m-1}(z) + K_m z^{-1} B_{m-1}(z) \quad , \quad m=1, 2, \dots, M-1$$

$$B_m(z) = z^{-m} A_m(z^{-1}) \quad , \quad m=1, 2, \dots, M-1$$

The soln is obtained recursively, beginning with $m=1$

e.g. Given a three-stage lattice filter with coeffs. $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{4}$, $K_3 = \frac{1}{3}$. Determine FIR filter coeff for the direct form structure.

Soln-
for
B. b)

Solⁿ: we have, $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{9}$, $K_3 = \frac{1}{3}$

we know,

$$A_m(z) = A_{m-1}(z) + K_m z^{-1} B_{m-1}(z)$$

For $m=1$

$$A_1(z) = A_0(z) + K_1 z^{-1} B_0(z)$$

$$= 1 + \frac{1}{4} z^{-1}$$

$$\because A_0(z) = B_0(z) = 1$$

$$\therefore A_1(z) = 1 + \frac{1}{4} z^{-1}$$

$$\therefore \alpha_1(0) = 1 \quad \& \quad \alpha_1(1) = K_1 = \frac{1}{4}$$

Since, $B_m(z)$ is reverse polynomial of $A_m(z)$, we have

$$B_1(z) = \frac{1}{4} + z^{-1}$$

For $m=2$

$$A_2(z) = A_1(z) + K_2 z^{-1} B_1(z)$$

$$= 1 + \frac{1}{4} z^{-1} + \frac{1}{9} \cdot z^{-1} \cdot \left(\frac{1}{4} + z^{-1} \right)$$

$$\therefore A_2(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2}$$

$$\therefore \alpha_2(0) = 1, \quad \alpha_2(1) = \frac{3}{2} \quad \& \quad \alpha_2(2) = K_2 = \frac{1}{2}$$

$$\& \quad B_2(z) = \frac{1}{2} + \frac{3}{8} z^{-1} + 1 \cdot z^{-2}$$

for $m=3$, $A_3(z) = A_2(z) + K_3 z^{-1} B_2(z)$

$$= 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{3} z^{-1} \left(\frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2} \right)$$

$$\therefore H(z) = A_3(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

$$\therefore \alpha_2(0) = 1, \alpha_2(1) = \frac{13}{24}, \alpha_2(2) = \frac{5}{8}, \alpha_2(3) = K_3 = \frac{1}{3}$$

Conversion of direct form FIR filter coeff. to lattice coeff.
we have

$$A_m(z) = A_{m-1}(z) + K_m z^{-1} B_{m-1}(z)$$

$$= A_{m-1}(z) + K_m [B_m(z) - K_m A_{m-1}(z)]$$

If we solve for $A_{m-1}(z)$, we obtain

$$A_{m-1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2}, \quad m = M-1, M-2, \dots, 1. \quad (a)$$

Thus, we compute all lower-degree polynomials $A_m(z)$ beginning with $A_{M-1}(z)$ and obtain the desired lattice coefficients from the relation $K_m = \alpha_m(m)$, if $|K_m| \neq 1$.

Beginning with $m = M-1$, in eqn (a), we have

$$K_m = \alpha_m(m), \quad \alpha_{m-1}(0) = 1$$

$$\therefore \alpha_{m-1}(k) = \frac{\alpha_m(k) - K_m \beta_m(k)}{1 - K_m^2}$$

$$\therefore \alpha_{m-1}(k) = \frac{\alpha_m(k) - \alpha_m(m) \alpha_m(m-k)}{1 - \alpha_m^2(m)}, \quad 1 \leq k \leq m-1$$

2011-fall 5.2) E.g. Determine the lattice coefs. corresponding to the FIR filter with system function.

$$H(z) = A_3(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}, \quad M=3.$$

Solⁿ: From above eqⁿ, $A_3(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$

we have, $\alpha_3(0)=1$, $\alpha_3(1) \neq \frac{13}{24}$, $\alpha_3(2) = \frac{5}{8}$, $\alpha_3 = K_3 = \frac{1}{3}$.

$$\therefore B_3(z) = \frac{1}{3} + \frac{5}{8} z^{-1} + \frac{13}{24} z^{-2} + z^{-3}.$$

we know,

$$A_{m-1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2}$$

for $m=3$,

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2} = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2}$$

For $m=2$,

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2} \quad \alpha_2(0)=1, \alpha_2(1) = \frac{3}{8}, \alpha_2(2) = \frac{1}{2} = K_2$$

$$\therefore B_2(z) = \frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2}$$

For $m=2$, $A_1(z) = \frac{A_2(z) - k_2 B_2(z)}{1 - k_2^2} = 1 + \frac{1}{4} z^{-1}$

$\alpha_1(0) = 1$ $\alpha_1(1) = k_1 = \frac{1}{4}$

$B_1(z) = \frac{1}{4} + z^{-1}$

Lattice and lattice-ladder structure for IIR systems

i) Lattice structure for IIR

Let an all pole system with system function

$$H(z) = \frac{1}{1 + \sum_{k=1}^N a_N z^{-k}} = \frac{1}{A_N(z)}$$

The difference equation for this IIR will be

$$y(n) = - \sum_{k=1}^N a_N(k) y(n-k) + x(n)$$

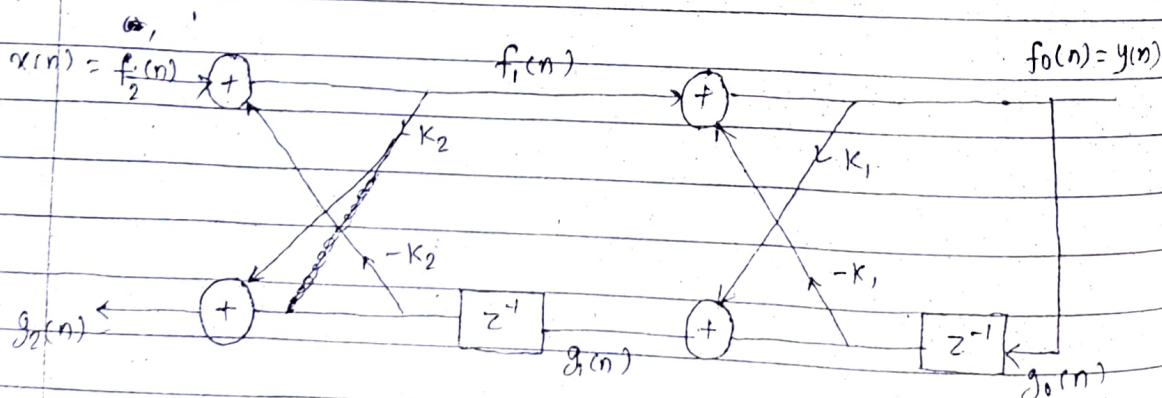


Fig: Two stage lattice structure of all pole IIR system

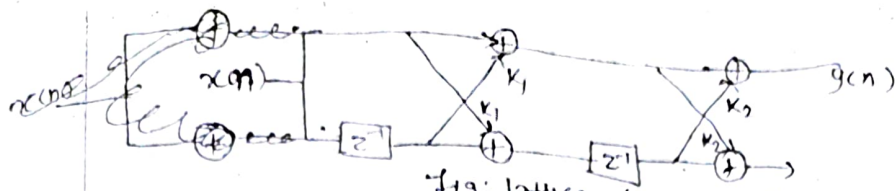


Fig: lattice structure for all zero system.

ii) lattice and ladder structure of IIR:

let us consider an IIR system, of

$$H(z) = \frac{B_M(z)}{A_M(z)} = \frac{\sum_{k=0}^N b_m(k) z^{-k}}{1 + \sum_{k=1}^N a_n(k) z^{-k}}, \quad N \geq M.$$

Algorithm:-

- i) find the lattice coeff. for all pole system and $A_m(z)$
draw the structure.
- ii) sort out the coeff. of $B_m(z)$ i.e. $b_0, b_1, b_2, \dots, b_n$.
- iii) Use $C_m = b_m = 1$, where, M is the order of polynomial.
for ladder
- iv) For other ladder coeff. use.

$$C_m = b_m - \sum_{i=m+1}^M C_i d_i (i-m), \quad m = M-1, M-2, \dots$$

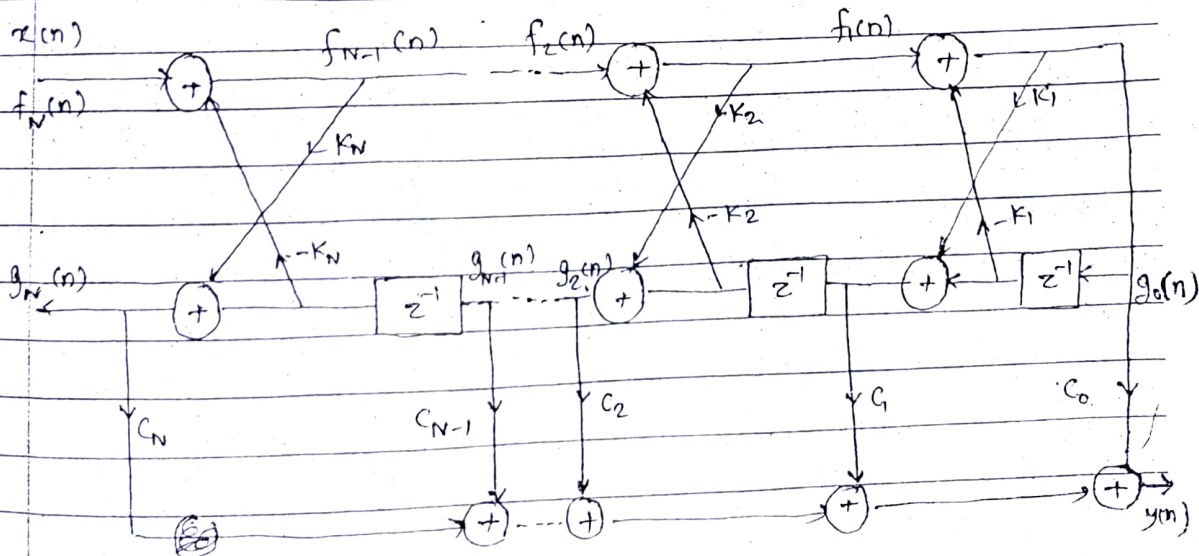


Fig: lattice-ladder structure for (IIR) all pole-zero system

Q. 13. (a) Obtain the direct form realization for the system described by the following equation.

$$y(n) = 0.5y(n-1) - 0.25y(n-2) + x(n) + 0.4x(n-1)$$

Solⁿ: Here given, difference equation,

$$y(n) = 0.5y(n-1) - 0.25y(n-2) + x(n) + 0.4x(n-1)$$

Taking z-transform on both sides, we get

$$Y(z) = 0.5z^{-1}Y(z) - 0.25z^{-2}Y(z) + X(z) + 0.4z^{-1}X(z)$$

$$Y(z) \{1 - 0.5z^{-1} + 0.25z^{-2}\} = \{0.4z^{-1} + 1\}X(z)$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 0.4z^{-1}}{1 - 0.5z^{-1} + 0.25z^{-2}} \quad \text{--- (1)}$$

From eqn (1), we observed that,

$$b_0 = 1, \quad b_1 = 0.4, \quad b_2 = 0$$

$$a_0 = 1, \quad a_1 = -0.5, \quad a_2 = +0.25$$

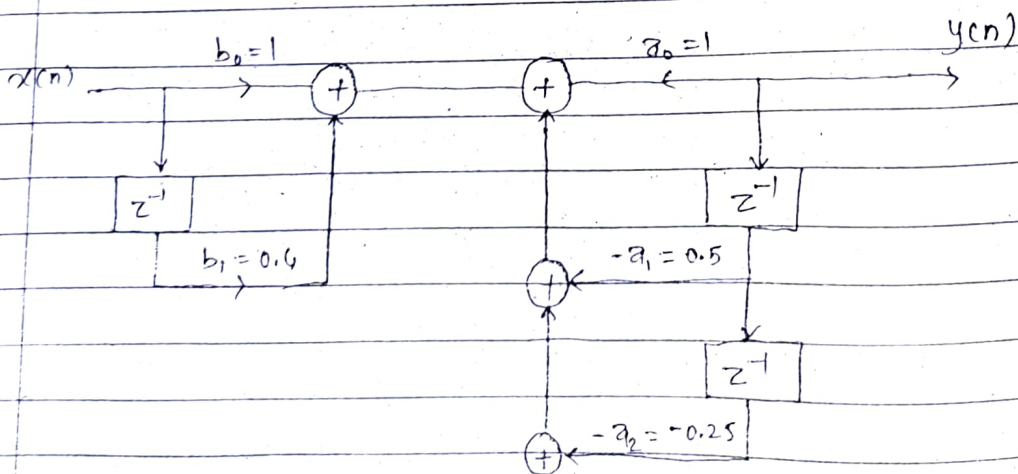


Fig: Direct form I Structure

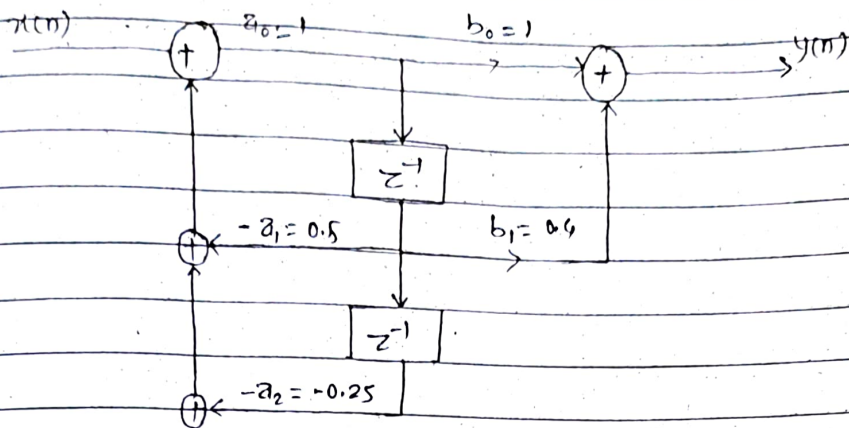


Fig: Direct form II structure

Q. obtain direct form I and II realization of a system described by

$$y(n) = \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{2}x(n-1)$$

Solⁿ - Here, given, difference, eqⁿ, is

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{2}x(n-1)$$

taking z-transform on both sides, we get

$$Y(z) - \frac{3}{4}z^{-1}Y(z) + \frac{1}{8}z^{-2}Y(z) = X(z) + \frac{1}{2}z^{-1}X(z)$$

$$\text{or, } Y(z) \left[1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2} \right] = X(z) \left[1 + \frac{1}{2}z^{-1} \right]$$

$$\text{or } H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}$$

$$b_0 = 1, \quad b_1 = \frac{1}{2}$$

$$a_0 = 1, \quad a_1 = -\frac{3}{4}, \quad a_2 = \frac{1}{8}$$

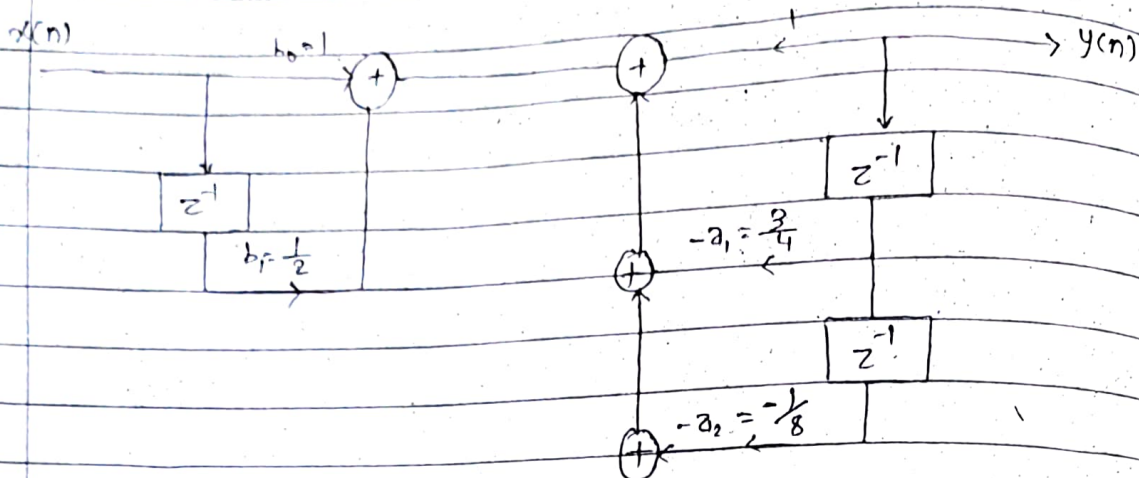


fig: Direct - form I structure

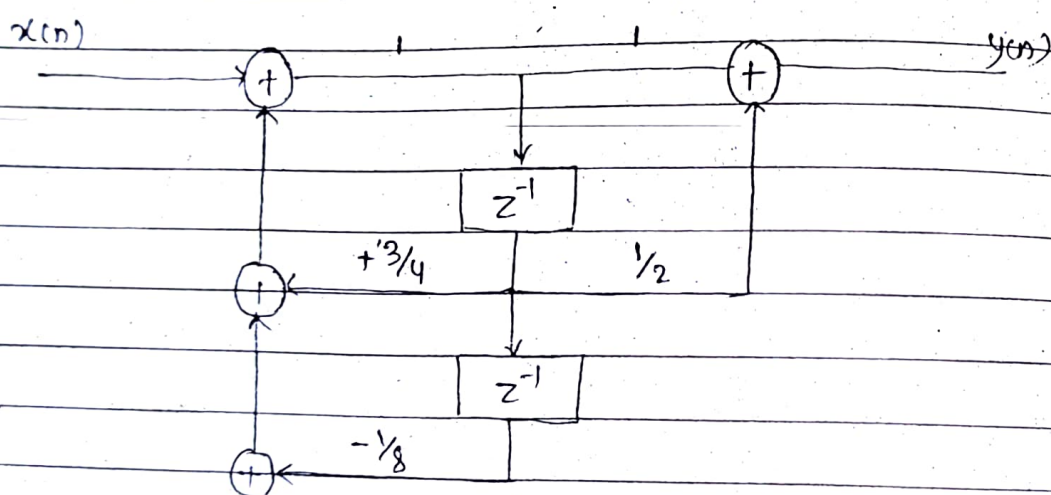


fig: Direct - form II structure

for cascade, $H(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}$ — (1)

let $H(z) = H_1(z) \cdot H_2(z)$

$\therefore$ from eqn (1), $H(z) = \frac{1 + \frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}$

$$\therefore H(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} \cdot \frac{1}{1 - \frac{1}{4}z^{-1}}$$

$\underbrace{\hspace{10em}}_{H_1(z)} \quad \underbrace{\hspace{10em}}_{H_2(z)}$

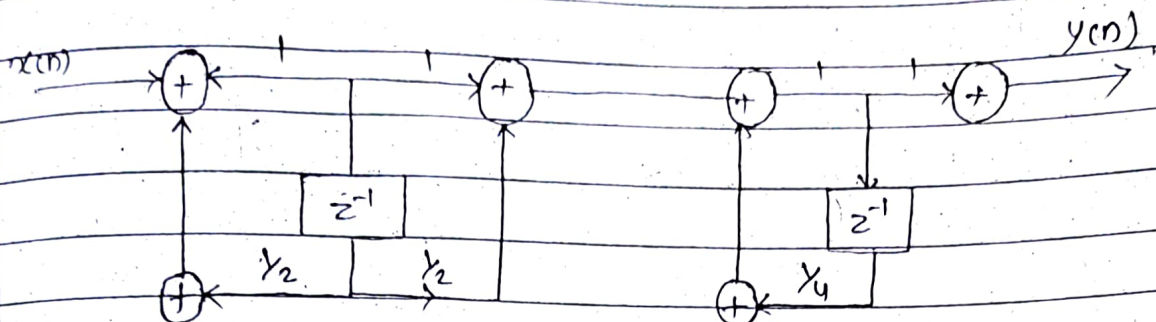


fig: Cascade Realization.

Q. Obtain the direct form realizations for the system described by the following equation

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$$

Solⁿ:

Here, given, difference equation, is

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$$

taking z-transform on both sides, we get

$$Y(z) - 3z^{-1}Y(z) - 4z^{-2}Y(z) = X(z) + 2z^{-1}X(z)$$

$$\therefore Y(z) \{1 - 3z^{-1} - 4z^{-2}\} = X(z) \{1 + 2z^{-1}\}$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 2z^{-1}}{1 - 3z^{-1} - 4z^{-2}}$$

$$b_0 = 1 \quad b_1 = 2$$

$$a_0 = 1 \quad a_1 = -3 \quad a_2 = -4$$

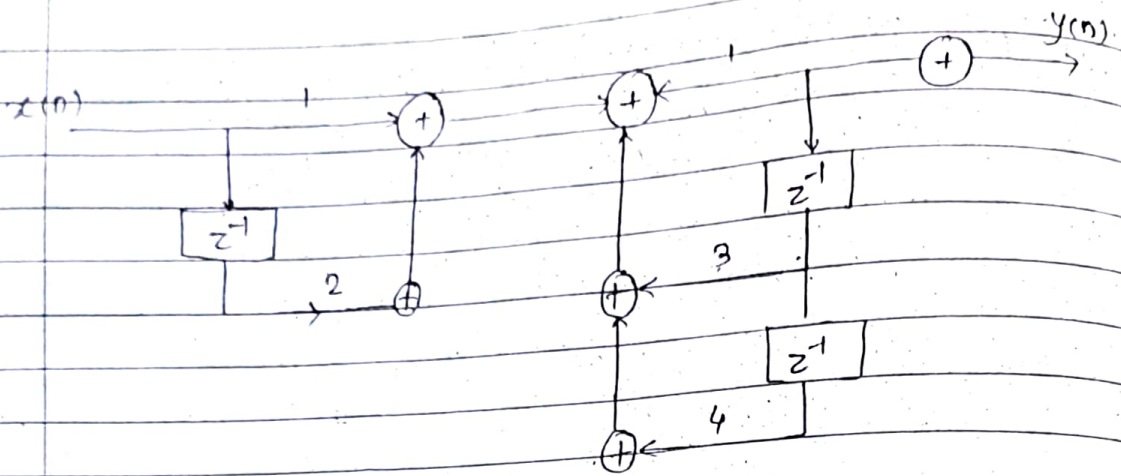


fig: Direct - form I structure

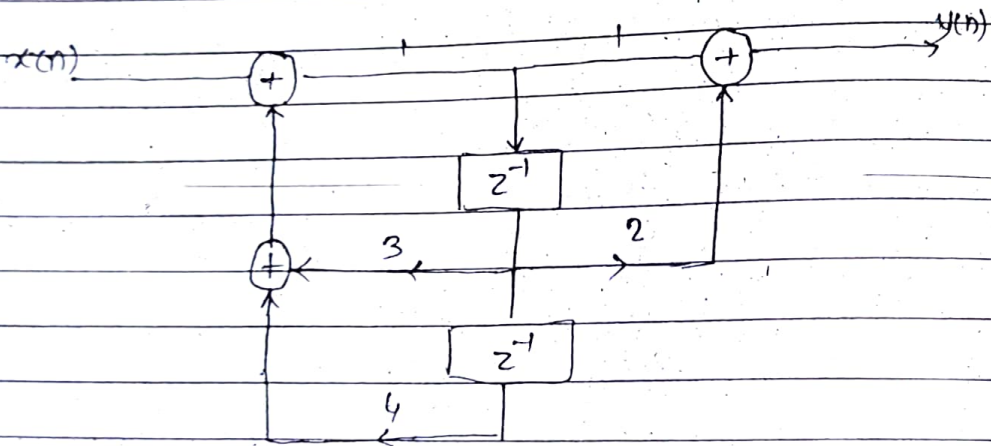


fig: Direct form II structure

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5.6)

Draw the cascade structure of following system.

$$y(n) = 2.3y(n-1) - 1.7y(n-2) + 0.4y(n-3) + x(n) - 2.3x(n-1)$$

Solⁿ: taking z-transform on both sides, we get

$$Y(z) = 2.3z^{-1}Y(z) - 1.7z^{-2}Y(z) + 0.4z^{-3}Y(z) + X(z) - 2.3z^{-1}X(z)$$

$$\text{or, } Y(z) \{ 1 - 2.3z^{-1} + 1.7z^{-2} - 0.4z^{-3} \} = X(z) \{ 1 - 2.3z^{-1} \}$$

$$\therefore H(z) = \frac{1 - 2.3z^{-1}}{1 - 2.3z^{-1} + 1.7z^{-2} - 0.4z^{-3}}$$

$$H(z) = \frac{1 - 2.3z^{-1}}{(1 - z^{-1})(1 - \frac{1}{2}z^{-1})(1 - \frac{4}{5}z^{-1})}$$

$$\therefore H(z) = \underbrace{\frac{1 - 2.3z^{-1}}{(1 - z^{-1})}}_{H_1(z)} \cdot \underbrace{\frac{1}{(1 - \frac{1}{2}z^{-1})}}_{H_2(z)} \cdot \underbrace{\frac{1}{(1 - \frac{4}{5}z^{-1})}}_{H_3(z)}$$

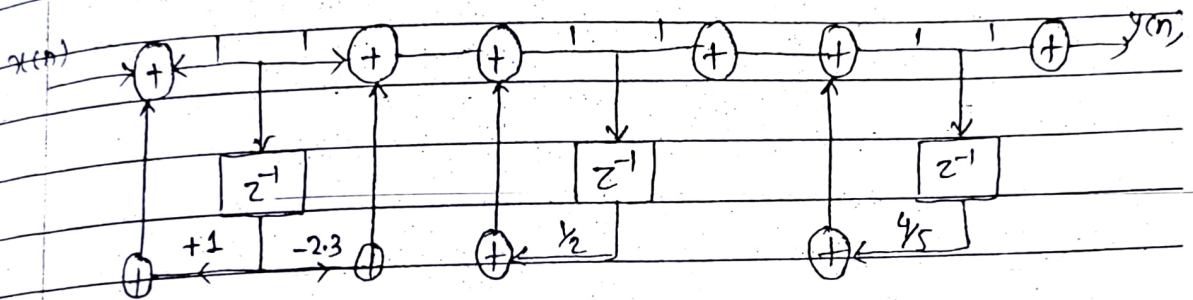


fig: Cascade structure.

Obtain cascade and parallel structure for the following system.

$$y(n) = y(n-1) - \frac{1}{2}y(n-2) + x(n) - x(n-1) + x(n-2)$$

Soln: Here, taking z-transform on both sides

$$Y(z) = z^{-1}Y(z) - \frac{1}{2}z^{-2}Y(z) + X(z) - z^{-1}X(z) + z^{-2}X(z)$$

$$\therefore Y(z) \left\{ 1 - z^{-1} + \frac{1}{2}z^{-2} \right\} = X(z) \left\{ 1 - z^{-1} + z^{-2} \right\}$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1 - z^{-1} + z^{-2}}{1 - z^{-1} + \frac{1}{2}z^{-2}}$$

For cascade structure,

$$H(z) = \frac{1 - z^{-1} + z^{-2}}{1 - z^{-1} + \frac{1}{2}z^{-2}}$$

2016-fall 5.2.8) Obtain the cascade form realization of following IIR filter, $H(z) = \frac{2(z+2)}{z(z-0.1)(z+0.5)(z-0.4)}$

Solⁿ:

$$\text{Here, } H(z) = \frac{2(z+2)}{z(z-0.1)(z+0.5)(z-0.4)}$$

$$= \frac{2z(1+2z^{-1})}{z^4(1-0.1z^{-1})(1+0.5z^{-1})(1-0.4z^{-1})}$$

$$= \frac{2z^{-3}(1+2z^{-1})}{(1-0.1z^{-1})(1+0.5z^{-1})(1-0.4z^{-1})}$$

$$\therefore H(z) = \underbrace{2z^{-3}}_{H_1(z)} \underbrace{(1+2z^{-1})}_{H_2(z)} \underbrace{(1+0.5z^{-1})}_{H_3(z)} \underbrace{(1-0.4z^{-1})}_{H_4(z)}$$

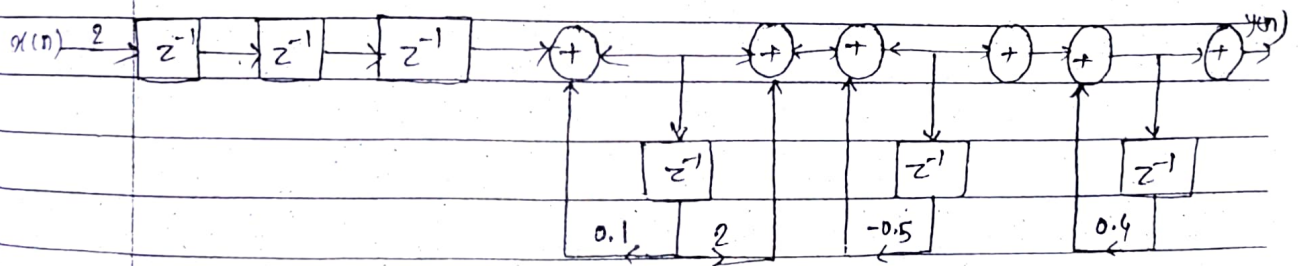


Fig: Cascade structure

2015-spring
4.6)

Obtain the parallel form realization of following IIR filter

$$H(z) = \frac{3(z^2 + 5z + 4)}{(z-1)(z+2)}$$

Solⁿ: Here, $H(z) = \frac{3(z^2 + 5z + 4)}{(2z+1)(z+2)}$

Let $\frac{H(z)}{z} = \frac{3(z^2 + 5z + 4)}{2z(2z+1)(z+2)}$ (i.e. dividing both sides by z)

Let By partial fraction method;

$$\frac{H(z)}{z} = \frac{A}{z} + \frac{B}{2z+\frac{1}{2}} + \frac{C}{z+2} \quad \text{--- (1)}$$

$$A = \frac{3(z^2 + 5z + 4)}{2(2z+1)(z+2)} \Big|_{z=0} = \frac{3 \cdot 4}{2 \cdot 1 \cdot 2} \Rightarrow A = \frac{3}{1} = 3$$

$$B = \frac{3(z^2 + 5z + 4)}{2z(z+2)} \Big|_{z=-\frac{1}{2}} = -8 \Rightarrow B = -8$$

$$C = \frac{3(z^2 + 5z + 4)}{z(2z+1)} \Big|_{z=-2} = 1 \Rightarrow C = 1$$

now,

eqⁿ (1), becomes

$$\frac{H(z)}{z} = \frac{3}{z} - \frac{8}{2z+\frac{1}{2}} + \frac{1}{z+2}$$

$$\therefore H(z) = 3 - \frac{4z}{z+\frac{1}{2}} + \frac{z}{z+2}$$

$$\therefore H(z) = 3 + \frac{(-4)}{1+\frac{1}{2}z^{-1}} + \frac{z}{1+2z^{-1}}$$

$$H(z) = H_1(z) + H_2(z) + H_3(z)$$

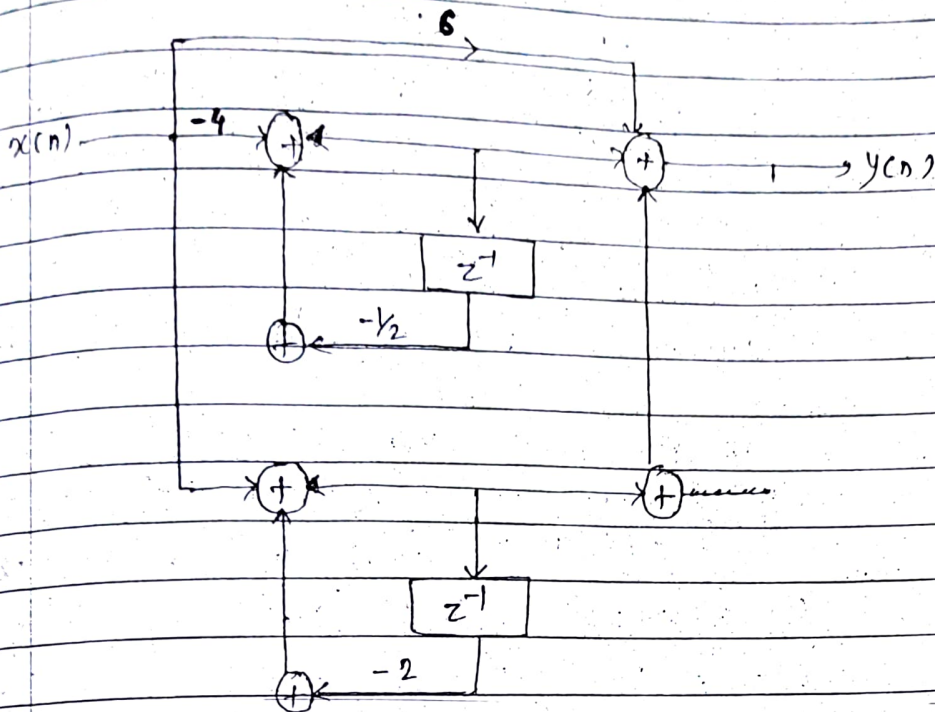


fig: parallel structure

Q. Draw the lattice structure of $H(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$
 soln: Let $A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$, order, $M=3$.

on solving we get, lattice coeff: $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{2}$, $K_3 = \frac{1}{3}$

Since the values of K_1, K_2 & K_3 are less than 1 so it is stable

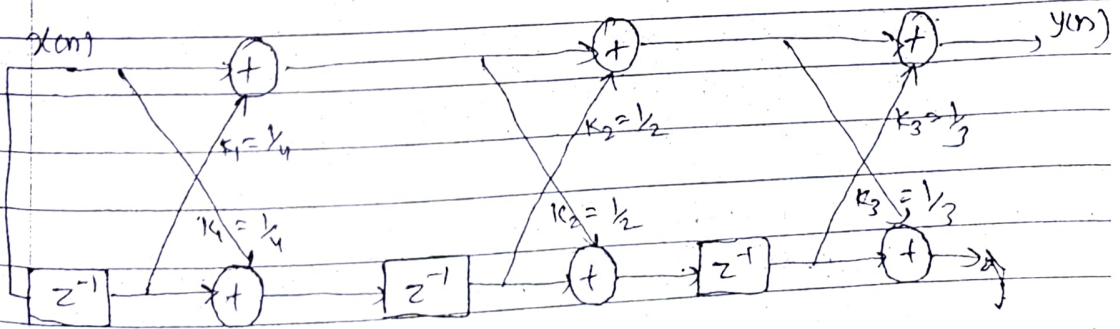


fig: lattice structure of all zero system. (no feed back)

Q Draw lattice structure of $H(z) = 2 + \frac{26}{24}z^{-1} + \frac{10}{8}z^{-2} + \frac{2}{3}z^{-3}$

i.e. $H(z) = 2 \left[1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3} \right] \rightarrow H(z) = 2A_3(z)$

on solving, we get $k_1 = \frac{1}{4}, k_2 = \frac{1}{2}, k_3 = \frac{1}{2}$

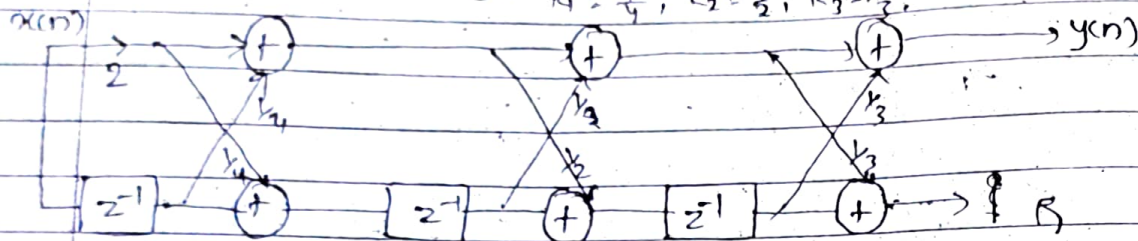


fig: lattice structure of all zero system (no feedback)

2011 Spring
5-2/

Q Draw lattice structure of IIR system. $H(z) = \frac{1}{2 + 1.4z^{-1} + 1.8z^{-2}}$

Soln. Let $H(z) = \frac{1}{A_2(z)}$ where $A_2(z) = 2 + 1.4z^{-1} + 1.8z^{-2} = 2(1 + 0.7z^{-1} + 0.9z^{-2})$

Let $A_2(z) = (1 + 0.7z^{-1} + 0.9z^{-2})$, $\alpha_2(0) = 1, \alpha_2(1) = 0.7, \alpha_2(2) = 0.9 = k_2$

we know, $A_{m-1}(z) = \frac{A_m(z) - k_m B_m(z)}{1 - k_m^2}$ & $B_2(z) = 0.9 + 0.7z^{-1} + z^{-2}$

for $m=2$

$$A_1(z) = \frac{A_2(z) - k_2 B_2(z)}{1 - k_2^2}$$

$$\therefore A_1(z) = \frac{1 + 0.7z^{-1} + 0.9z^{-2} - 0.9(0.9 + 0.7z^{-1} + z^{-2})}{1 - (0.9)^2}$$

$$= \frac{0.19 + 0.07z^{-1} + 0}{0.19}$$

$$\therefore A_1(z) = 1 + 0.368z^{-1}$$

we have $A_1(z) = 1 + 0.368z^{-1}$

$$B_1(z) = 0.368 + z^{-1}$$

$$\alpha_1(0) = 1, \alpha_1(1) = 0.368 = k_1$$

Star

$\therefore$ lattice coeff. are $k_1 = 0.368$ & $k_2 = 0.9$, which are less than 1, so it is stable system

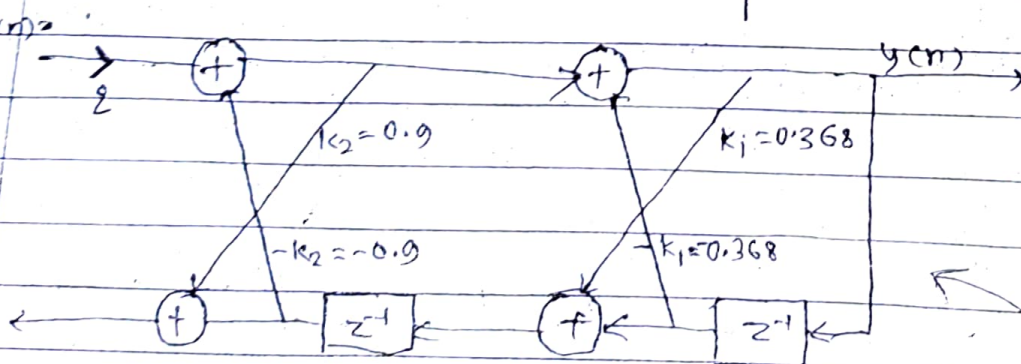
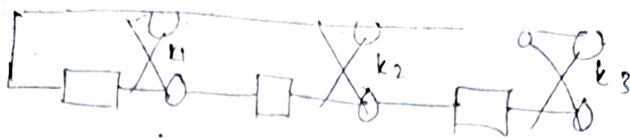


fig: lattice structure of all pole system (feedback)



Q. 11 - 12

Realize the IIR filter $H(z)$ using Direct form II and lattice structure.

$$H(z) = \frac{1}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}} \quad \left| \begin{array}{l} \text{All} \\ \text{poles} \end{array} \right.$$

Solⁿ: $b_0 = 1, a_0 = 1, a_1 = \frac{13}{24}, a_2 = \frac{5}{8}, a_3 = \frac{1}{3}$

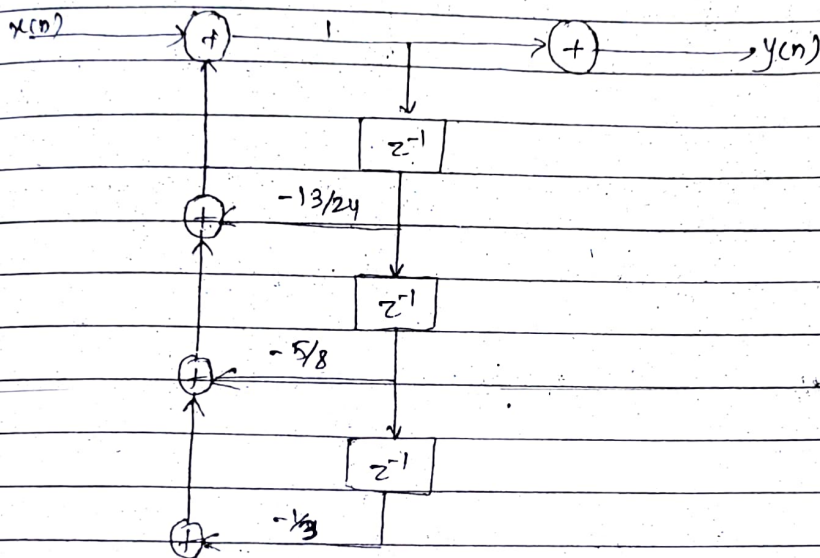


Fig: Direct form II structure for $H(z)$

Here, $H(z) = \frac{1}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$

Let $H(z) = \frac{1}{A_3(z)}$

$\therefore A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$

$\alpha_3(0) = 1, \alpha_3(1) = \frac{13}{24}, \alpha_3(2) = \frac{5}{8}, \alpha_3(3) = \frac{1}{3}$

$B_3(z) = \frac{1}{3} + \frac{5}{8}z^{-1} + \frac{13}{24}z^{-2} + z^{-3}$

we know,

$$A_{m-1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2}$$

for $m=3$

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$\text{or, } A_2(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{4}z^{-3} - \frac{1}{4} \left[\frac{1}{4} + \frac{5}{8}z^{-1} + \frac{13}{24}z^{-2} + z^{-3} \right]$$

~~Ans is wrong~~

$$= \frac{15}{16} + \frac{37}{96}z^{-1} + \frac{47}{96}z^{-2} - \frac{1}{16}$$

~~must be $\frac{1}{3}$~~

~~$\frac{1}{4}$~~

$$\therefore A_2(z) = 5 + \frac{37}{18}z^{-1} + \frac{47}{18}z^{-2}$$

$$\& B_2(z) = \frac{47}{18} + \frac{37}{18}z^{-1} + 5z^{-2}$$

$$\alpha_2(0) = 5 \quad \alpha_2(1) = \frac{37}{18} \quad , \quad \alpha_2(2) = \frac{47}{18} = K_2$$

for $m=2$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$\text{or } A_1(z) = \frac{5 + \frac{37}{18}z^{-1} + \frac{47}{18}z^{-2} - \frac{47}{18} \left(\frac{47}{18} + \frac{37}{18}z^{-1} + 5z^{-2} \right)}{1 - \left(\frac{47}{18} \right)^2}$$

$$A_0(z) =$$

we know,

$$A_{m+1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2}$$

for $m=3$

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$A_2(z) = \frac{1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3} - \frac{1}{3} \left[\frac{1}{3} + \frac{5}{8} z^{-1} + \frac{13}{24} z^{-2} + z^{-3} \right]}{1 - \left(\frac{1}{3}\right)^2}$$

$$\therefore A_2(z) = \frac{\frac{8}{9} + \frac{1}{3} z^{-1} + \frac{4}{9} z^{-2}}{\frac{8}{9}} = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2}$$

$$\alpha_2(0) = 1 \quad \alpha_2(1) = \frac{3}{8} \quad \alpha_2(2) = K_2 = \frac{1}{2}$$

$$B_2(z) = \frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2}$$

for $m=2$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2} = 1 + \frac{1}{4} z^{-1}$$

$$\alpha_1(0) = 1 \quad \alpha_1(1) = K_1 = \frac{1}{4}$$

$$B_1(z) = \frac{1}{4} + z^{-1}$$

$\therefore$ The lattice coeffs are $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{2}$, $K_3 = \frac{1}{3}$.

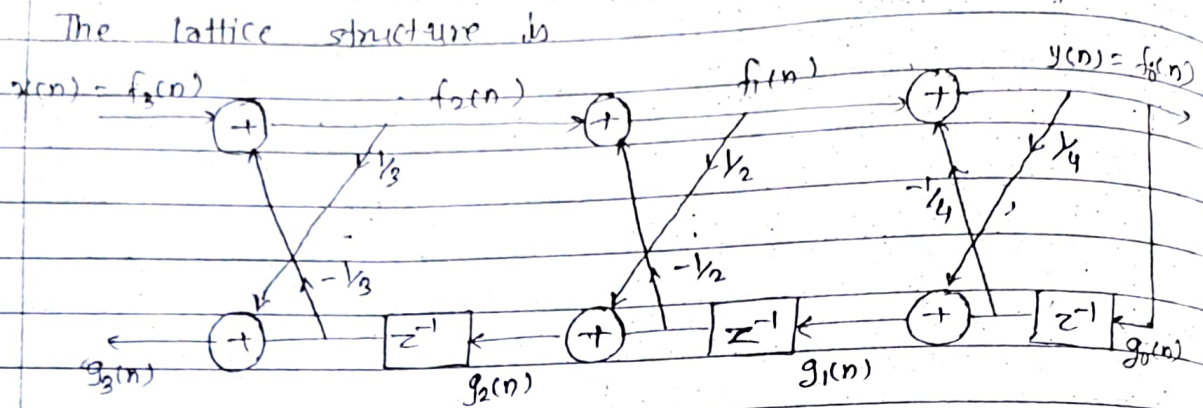


fig: lattice structure of all pole polynomial.

2019-fall
5.2)

Draw the lattice structure of the following system

$$H(z) = \frac{2}{1 - 0.5z^{-1} + 0.8z^{-2} + 0.9z^{-3}}$$

Solⁿ: we know, $H(z) = \frac{1}{A_m(z)}$

$$\therefore H(z) = \frac{1}{\frac{1}{2} (1 - 0.5z^{-1} + 0.8z^{-2} + 0.9z^{-3})}$$

let, $A_3(z) = 1 - 0.5z^{-1} + 0.8z^{-2} + 0.9z^{-3}$

$\alpha_3(0) = 1$ $\alpha_3(1) = -0.5$ $\alpha_3(2) = 0.8$ $\alpha_3(3) = 0.9$

$B_3(z) = 0.9 + 0.8z^{-1} - 0.5z^{-2} + z^{-3}$

we know,

$$A_{m+1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2}$$

for $m=3$

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$A_2(z) = \frac{1 - 0.5z^{-1} + 0.8z^{-2} + 0.9z^3 - 0.9(0.9 + 0.8z^{-1} - 0.5z^{-2} + z^{-3})}{1 - (0.9)^2}$$

$$= \frac{0.19 - 1.22z^{-1} + 1.25z^{-2} + 0}{0.19}$$

$$\therefore A_2(z) = 1 - 6.42z^{-1} + 6.57z^{-2}$$

$$\alpha_2(0) = 1 \quad \alpha_2(1) = -6.54 \quad \alpha_2(2) = 6.57 = K_2$$

$$B_2(z) = 6.57 - 6.42z^{-1} + z^{-2}$$

for $m=2$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$A_1(z) = \frac{1 - 6.42z^{-1} + 6.57z^{-2} - 6.57(6.57 - 6.42z^{-1} + z^{-2})}{1 - (6.57)^2}$$

$$A_1(z) = \frac{-42.165 + 37.86z^{-1} + 0}{-42.165} = 1 - 0.895z^{-1}$$

$$\alpha_1(0) = 1 \quad \alpha_1(1) = -0.895 = K_1$$

$$B_1(z) = -0.895 + z^{-1}$$

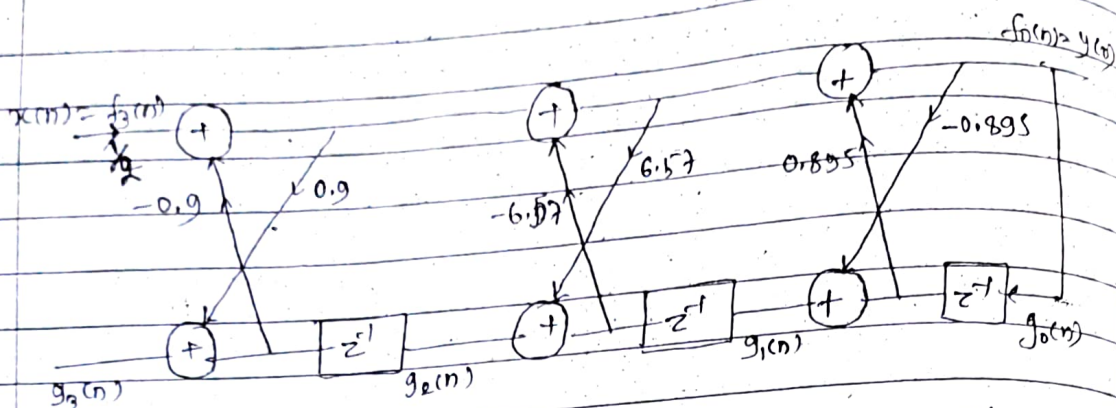


fig: lattice structure of all pole polynomial.

2012- Fall
4. b)

Q. Convert the following pole zero IIR filter into the lattice ladder structure.

$$H(z) = \frac{1 + 2z^{-1} + 2z^{-2} + z^{-3}}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}} \rightarrow \text{gives ladder}$$

$$L(1)$$

Soln:-

$$\text{Let } H(z) = \frac{B_3(z)}{A_3(z)} \quad \text{--- (1)}$$

Comparing eqn (1) & (11),

For lattice coeff of all pole system,

$$A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$$

$$\alpha_3(0) = 1, \alpha_3(1) = \frac{13}{24}, \alpha_3(2) = \frac{5}{8}, \alpha_3(3) = k_3 = \frac{1}{3}$$

$$\& B_3(z) = \frac{1}{3} + \frac{8}{3}z^{-1} + \frac{13}{24}z^{-2} + z^{-3}$$

we know,

$$A_{m-1}(z) = \frac{A_m(z) - k_m B_m(z)}{1 - k_m^2}$$

for $m=3$

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$A_2(z) = \frac{1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3} - \frac{1}{3} \left(\frac{1}{3} + \frac{5}{8} z^{-1} + \frac{13}{24} z^{-2} + z^{-3} \right)}{1 - \left(\frac{1}{3} \right)^2}$$

$$\therefore A_2(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2}$$

$$\alpha_2(0) = 1, \quad \alpha_2(1) = \frac{3}{8}, \quad \alpha_2(2) = K_2 = \frac{1}{2},$$

$$B_2(z) = \frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2}$$

for $m=2$,

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$A_1(z) = \frac{1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2} - \frac{1}{2} \left(\frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2} \right)}{1 - \left(\frac{1}{2} \right)^2}$$

$$\therefore A_1(z) = 1 + \frac{1}{4} z^{-1}$$

$$\alpha_1(0) = 1, \quad \alpha_1(1) = K_1 = \frac{1}{4}$$

$\therefore$ lattice coeff. are $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{2}$, & $K_3 = \frac{1}{3}$

for ladder coeff. of all zero system. i.e

$$B_3(z) = 1 + 2z^{-1} + 2z^{-2} + z^{-3}, \quad M = 3. \text{ (highest -ve power)}$$

$$b_0 = 1 \quad b_1 = 2 \quad b_2 = 2 \quad b_3 = 1$$

we know, $c_3 = b_3 = 1$.

For other ladder coeff.

$$c_m = b_m - \sum_{i=m+1}^M c_i d_i (i-m)$$

for $m=2$

$$c_2 = b_2 - \sum_{i=3}^3 c_i d_i (i-2)$$

$$= b_2 - c_3 d_3 (3-2)$$

$$= b_2 - c_3 d_3 (1)$$

$$= 2 - 1 \cdot \frac{13}{24}$$

$$\therefore c_2 = \frac{35}{24}$$

for $m=1$

$$c_1 = b_1 - \sum_{i=2}^3 c_i d_i (i-1)$$

$$= b_1 - c_2 d_2 (2-1) - c_3 d_3 (3-1)$$

$$= b_1 - c_2 d_2 (1) - c_3 d_3 (2)$$

$$= 2 - \frac{35}{24} \times \frac{3}{8} - 1 \times \frac{5}{8}$$

$$= 52$$

∴ ladder coeffs are $c_1 = \frac{53}{64}$, $c_2 = \frac{35}{24}$, & $c_3 = 1$.

for $m=0$

$$c_0 = b_0 - \sum_{i=1}^3 c_i \alpha_i (i-0)$$

$$= 1 - c_1 \alpha_1 (1) - c_2 \alpha_2 (2) - c_3 \alpha_3 (3)$$

$$= 1 - \frac{53}{64} \times \frac{1}{4} - \frac{35}{24} \times \frac{1}{2} - 1 \times \frac{1}{3}$$

$$\therefore c_0 = -0.2695$$

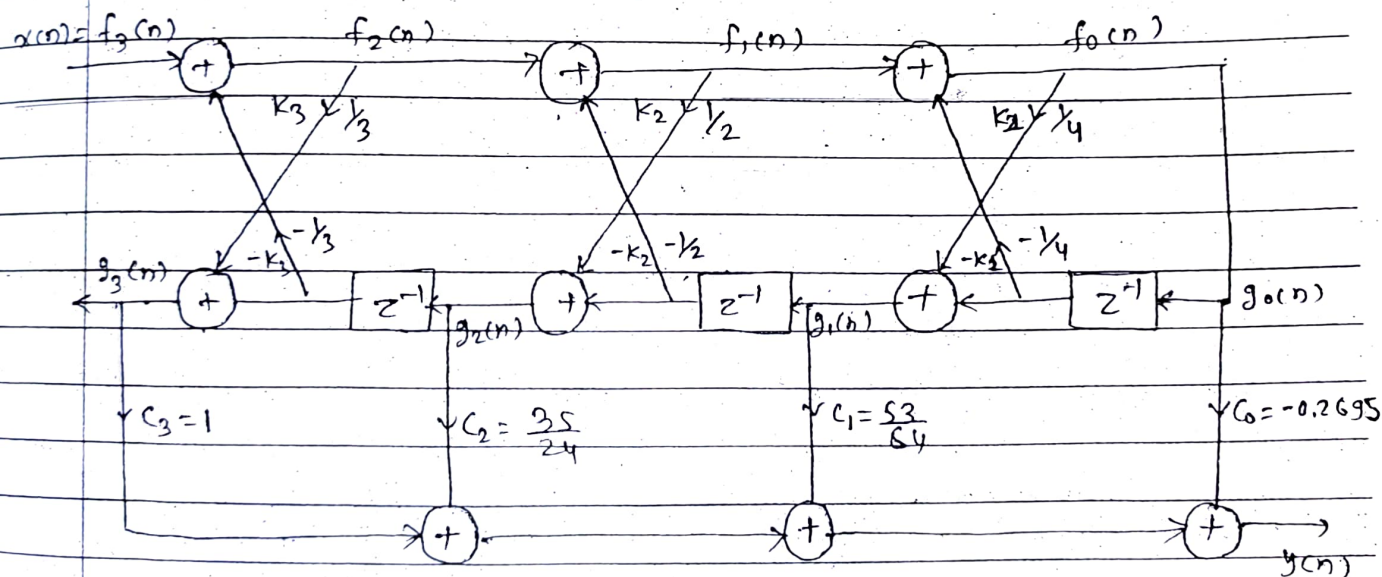


Fig: Lattice-ladder structure for pole-zero polynomial

Convert the following IIR filter into lattice ladder structure

$$H(z) = 1 + z^{-1} + 2z^{-2} + z^{-3}$$

$$1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

Solⁿ for lattice coeff. of all pole system

$$A_3(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

$$\alpha_3(0) = 1 \quad \alpha_3(1) = \frac{13}{24}, \quad \alpha_3(2) = \frac{5}{8}, \quad \alpha_3(3) = \frac{1}{3} = k_3$$

using formula,

$$A_{m-1}(z) = \frac{A_m(z) - k_m B_m(z)}{1 - k_m^2}$$

$$\text{For } m=3, \quad A_2(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2}$$

$$\alpha_2(0) = 1 \quad \alpha_2(1) = \frac{3}{8}, \quad \alpha_2(2) = k_2 = \frac{1}{2}$$

$$B_2(z) = \frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2}$$

$$\text{for } m=2 \quad A_1(z) = 1 + \frac{1}{4} z^{-1}$$

$$\alpha_1(0) = 1 \quad \alpha_1(1) = \frac{1}{4} = k_1$$

for ladder coeff. of all zero system

$$B_3(z) = 1 + z^{-1} + 2z^{-2} + z^{-3}, \quad M=3$$

$$b_0 = 1, \quad b_1 = 1, \quad b_2 = 2, \quad b_3 = c_3 = 1$$

$$\text{We know, } c_m = b_m - \sum_{i=m+1}^M c_i \alpha_i(i-m)$$

for $m=2$

$$c_2 = b_2 - \sum_{i=3}^3 c_i \alpha_i(i-2)$$

$$c_2 = 2 - c_3 d_3(1) = 2 - 1 \times \frac{13}{24} = \frac{35}{24}$$

for $m=1$

$$c_1 = b_1 - \sum_{i=2}^3 c_i d_i(1)$$

$$= 1 - c_2 d_2(1) - c_3 d_3(2)$$

$$= 1 - \frac{35}{24} \times \frac{3}{8} - 1 \times \frac{5}{8}$$

$$\therefore c_1 = -\frac{11}{64} = -0.1718$$

for $m=0$

$$c_0 = b_0 - \sum_{i=1}^3 c_i d_i(1)$$

$$= 1 - c_1 d_1(1) - c_2 d_2(2) - c_3 d_3(3)$$

$$= 1 - \left(-\frac{11}{64}\right) \times \frac{1}{4} - \frac{35}{24} \times \frac{1}{2} - 1 \times \frac{1}{3}$$

$$= -\frac{5}{256}$$

$$\therefore c_0 = -0.0195$$

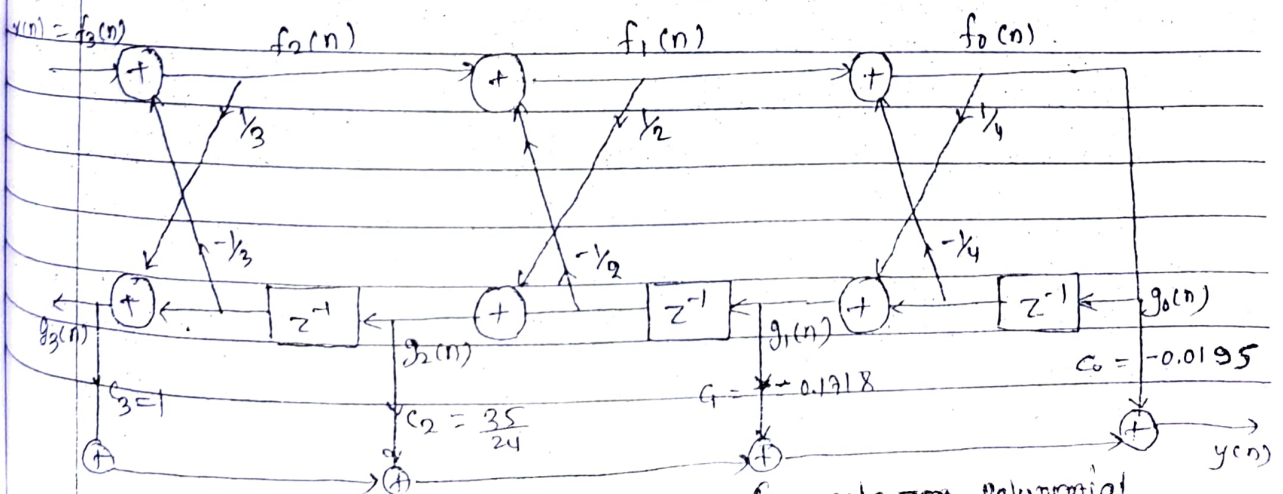


Fig lattice - lattice structure for pole-zero polynomial

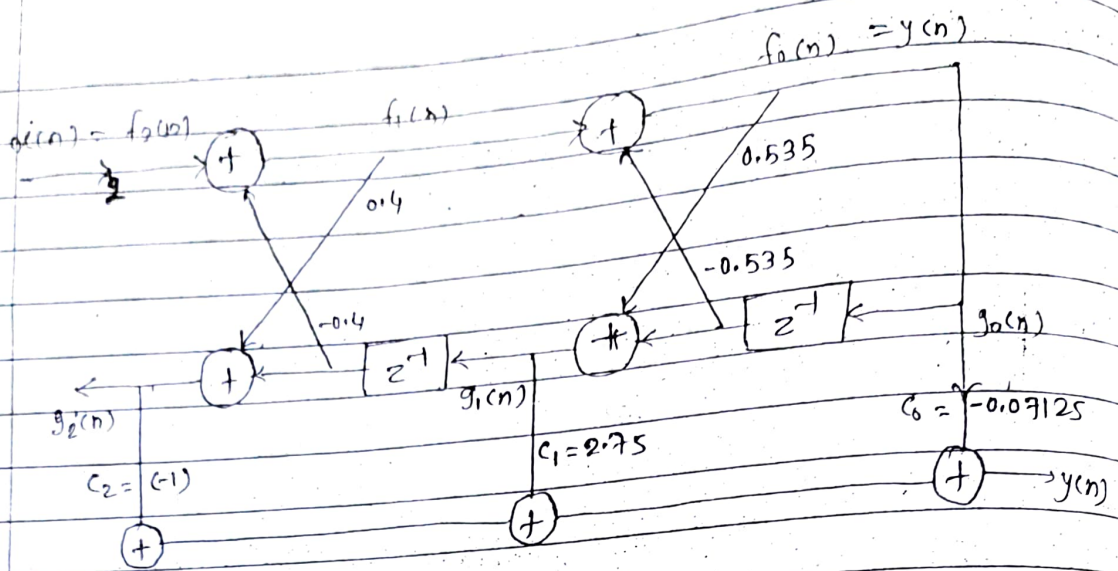


fig: lattice-ladder structure of ~~the~~ pole-zero system

Q. 4.6) Spring

Draw the lattice-ladder structure for the following system.

$$H(z) = \frac{1 - 0.85z^{-1} + 0.15z^{-2}}{1 + 0.44z^{-1} - 0.82z^{-2}}$$

Soln:-

$$\text{Let } H(z) = \frac{B_m(z)}{A_m(z)}$$

for lattice coeff

$$\text{Let } A_2(z) = 1 + 0.44z^{-1} - 0.82z^{-2}$$

$$\alpha_2(0) = 1, \quad \alpha_2(1) = 0.44, \quad \alpha_2(2) = -0.82 = k_2$$

$$B_2(z) = -0.82 + 0.44z^{-1} + z^{-2}$$

we know, $A_{m-1}(z) = \frac{A_m(z) - k_m B_m(z)}{1 - k_m^2}$

for $m=2$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$\begin{aligned} A_1(z) &= \frac{1 + 0.44z^{-1} - 0.82z^{-2} + (-0.82)(-0.82 + 0.44z^{-1} + z^{-2})}{1 - (-0.82)^2} \\ &= \frac{0.3276 + 0.8008z^{-1}}{0.3276} \end{aligned}$$

$$\therefore A_1(z) = 1 + 2.44z^{-1}$$

$$\alpha_1(0) = 1 \quad \alpha_1(1) = 2.44 = K_1$$

$\therefore$ lattice coeff are $K_1 = 2.44$ & $K_2 = -0.82$

for ladder coeff.

$$\text{let } B_2(z) = 1 - 0.85z^{-1} + 0.15z^{-2}, \quad M=2$$

$$b_0 = 1 \quad b_1 = -0.85 \quad b_2 = 0.15 = c_2$$

we know,

$$c_m = b_m - \sum_{i=m+1}^M c_i \alpha_i(i-m)$$

for $m=1$

$$c_1 = b_1 - \sum_{i=2}^2 c_i \alpha_i(i-1)$$

$$= -0.85 - c_2 \alpha_2(1)$$

$$= -0.85 - 0.15 \times 0.44$$

$$\therefore c_1 = -0.916$$

For $m=0$

$$C_0 = b_0 - \sum_{i=1}^2 c_i d_i (i-1)$$

$$= b_0 - C_1 d_1 (1) - C_2 d_2 (2)$$

$$= 1 - (-0.916) \times 2.44 - (-0.82) \times 0.15$$

$$\therefore C_0 = 3.35804$$

$\therefore$ ladder coeff $C_0 = 3.358$ $C_1 = -0.916$ $C_2 = 0.15$

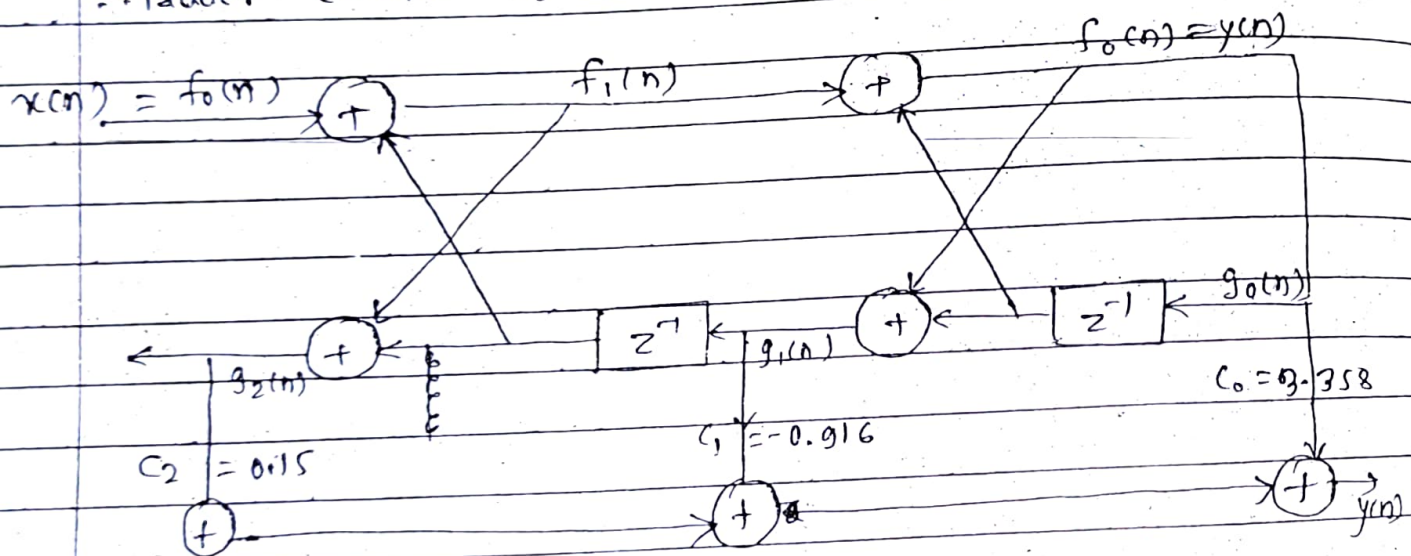


Fig: lattice-ladder structure of pole-zero system

2014

An LTI system is characterized by the system function

$$H(z) = (3 - 4z^{-1})$$

$$\left(\frac{3}{2} z^1 + \frac{3}{2} z^{-2} \right)$$

for $m=2$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$A_1(z) = \frac{1 + 0.44z^{-1} - 0.82z^{-2} + (0.82)(-0.82 + 0.44z^{-1} + z^{-2})}{1 - (-0.82)^2}$$

$$= \frac{0.3276 + 0.8008z^{-1}}{0.3276}$$

$$\therefore A_1(z) = 1 + 2.44z^{-1}$$

$$\alpha_1(0) = 1 \quad \alpha_1(1) = 2.44 = K_1$$

$\therefore$ lattice coeff are $K_1 = 2.44$ & $K_2 = -0.82$

for ladder coeff.

$$\text{let } B_2(z) = 1 - 0.85z^{-1} + 0.15z^{-2}, \quad M=2$$

$$b_0 = 1 \quad b_1 = -0.85 \quad b_2 = 0.15 = c_2$$

we know,

$$c_m = b_m - \sum_{i=m+1}^M c_i \alpha_i(i-m)$$

for $m=1$

$$c_1 = b_1 - \sum_{i=2}^2 c_i \alpha_i(i-1)$$

$$= -0.85 - c_2 \alpha_2(1)$$

$$= -0.85 - 0.15 \times 0.44$$

$$\therefore c_1 = -0.916$$

For $m=0$

$$C_0 = b_0 - \sum_{i=1}^2 c_i \alpha_i (i-0)$$

$$= b_0 - C_1 \alpha_1 (1) - C_2 \alpha_2 (2)$$

$$= 1 - (-0.916) \times 2.44 - (-0.82) \times 0.15$$

$$\therefore C_0 = 3.35804$$

$\therefore$ ladder coeff $C_0 = 3.358$ $C_1 = -0.916$ $C_2 = 0.15$

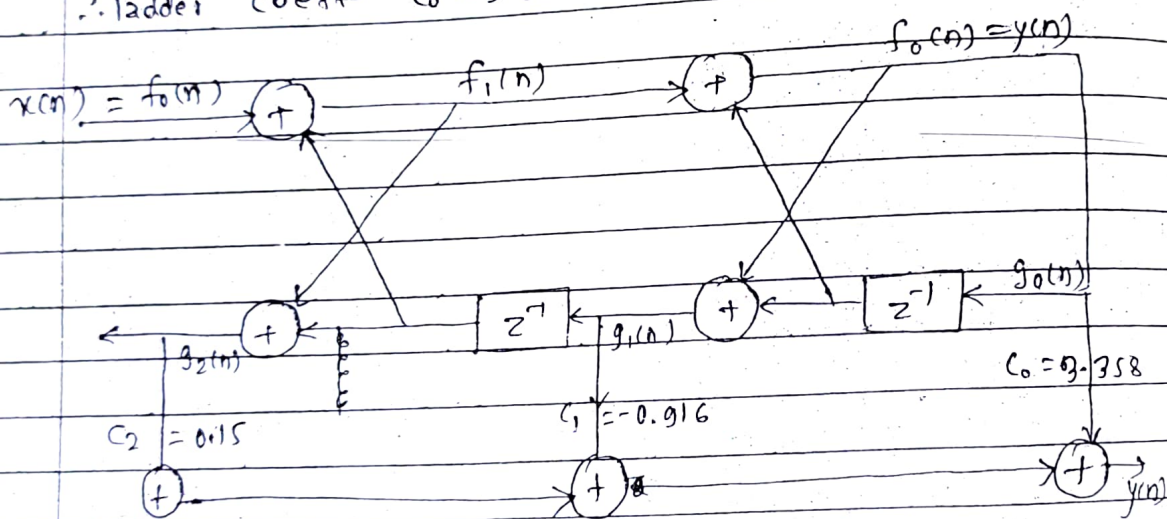


Fig: lattice-ladder structure of pole-zero system

Q14

An LTI system is characterized by the system function

$$H(z) = (3 - 4z^{-1})$$

$$\left(1 - \frac{4}{3}z^{-1} + \frac{2}{3}z^{-2}\right)$$

convert the following pole zero lattice filter into a lattice ladder structure.

$$H(z) = \frac{(1 + \frac{4}{3}z^{-1} - \frac{1}{3}z^{-2})}{(1 - \frac{3}{4}z^{-1})(1 - z^{-1} + \frac{1}{2}z^{-2})}$$

So

$$H(z) = \frac{1 + \frac{4}{3}z^{-1} - \frac{1}{3}z^{-2}}{1 - z^{-1} + \frac{1}{2}z^{-2} - \frac{3}{4}z^{-1} + \frac{3}{4}z^{-2} - \frac{3}{8}z^{-3}}$$

$$H(z) = \frac{1 + \frac{4}{3}z^{-1} - \frac{1}{3}z^{-2}}{1 - \frac{7}{4}z^{-1} + \frac{5}{4}z^{-2} - \frac{3}{8}z^{-3}} \quad \begin{matrix} Y(z) \\ X(z) \end{matrix}$$

Let $H(z) = \frac{B_m(z)}{A_m(z)}$. Therefore, for lattice coeffs.:

Also, let $A_3(z) = A_m(z) = 1 - \frac{7}{4}z^{-1} + \frac{5}{4}z^{-2} - \frac{3}{8}z^{-3}$

$\alpha_3(0) = 1, \alpha_3(1) = -\frac{7}{4}, \alpha_3(2) = \frac{5}{4}, \alpha_3(3) = -\frac{3}{8} = K_3$

we know, $B_3(z) = -\frac{3}{8} + \frac{5}{4}z^{-1} - \frac{7}{4}z^{-2} + z^{-3}$.

$$A_{m-1}(z) = A_m(z) - K_m B_m(z)$$

$$1 - K_m^2$$

for $m=3$

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$A_2(z) = \frac{1 - \frac{7}{4}z^{-1} + \frac{5}{4}z^{-2} - \frac{3}{8}z^{-3} + (\frac{3}{8}) \left[-\frac{3}{8} + \frac{5}{4}z^{-1} - \frac{7}{4}z^{-2} + z^{-3} \right]}{1 - (\frac{3}{8})^2}$$

$$= \frac{\frac{55}{64} - \frac{41}{32}z^{-1} + \frac{19}{32}z^{-2}}{\frac{55}{64}}$$

$$\therefore A_2(z) = 1 - \frac{82}{55} z^{-1} + \frac{38}{55} z^{-2} = 1 - 1.49 z^{-1} + 0.69 z^{-2}$$

$$\alpha_2(0) = 1 \quad \alpha_2(1) = -1.49 \quad \alpha_2(2) = 0.69 = K_2$$

$$B_2(z) = 0.69 - 1.49 z^{-1} + z^{-3}$$

for $m=2$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$\therefore A_1(z) = \frac{1 - 1.49 z^{-1} + 0.69 z^{-2} - 0.69(0.69 - 1.49 z^{-1} + z^{-3})}{1 - (0.69)^2}$$

$$\therefore A_1(z) = \frac{0.5239 - 0.4619 z^{-1} + 0}{0.5239}$$

$$\therefore A_1(z) = 1 - 0.88 z^{-1}$$

$$\alpha_1(0) = 1 \quad \alpha_1(1) = -0.88$$

for ladder Coeff.

$$B_2(z) = 1 + \frac{4}{3} z^{-1} - \frac{1}{3} z^{-2}, \quad M=2$$

$$b_0 = 1 \quad b_1 = \frac{4}{3} = 1.33 \quad b_2 = -\frac{1}{3} = -0.33 = G_2$$

$$\text{we know, } c_m = b_m - \sum_{i=m+1}^M c_i d_i(i-m)$$

for $m=1$

$$c_1 = b_1 - \sum_{i=2}^2 c_i d_i(i-1)$$

$$= b_1 - c_2 d_2(1)$$

$$= 1.33 - (-0.33) \times (-1.49)$$

$$\therefore c_1 = 0.83$$

for $m=0$,

$$C_0 = b_0 - \sum_{i=1}^2 C_i d_i (1-0)$$

$$= b_0 - C_1 d_1 (1) - C_2 d_2 (2)$$

$$= 1 - 0.83 \times (-0.88) - (-0.33) \times (0.69)$$

$$\therefore C_0 = 1.9581$$

$\therefore$ ladder coeff are, $C_0 = 1.9581$, $C_1 = 0.83$, $C_2 = -0.33$.

Lattice coeff are, $k_1 = -0.88$, $k_2 = 0.69$, $k_3 = -\frac{3}{8} = -0.375$

Since, all k_1, k_2 , & k_3 are less than 1 so, it is stable system.

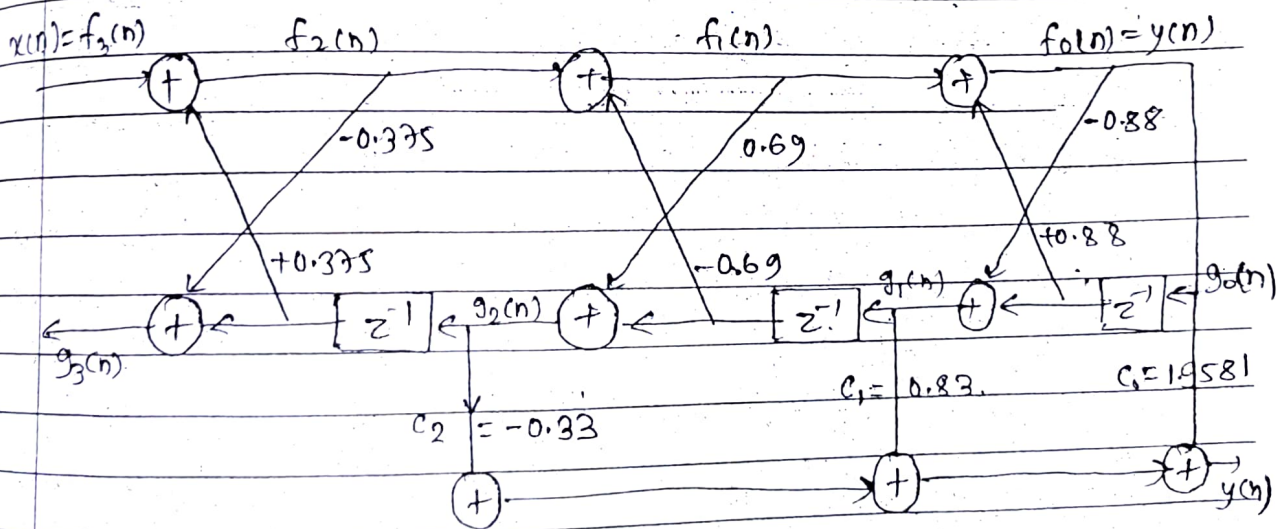


Fig: lattice-ladder structure of pole-zero system

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4.6)

Q. Draw the lattice ladder structure for the following IIR system.

$$H(z) = \frac{1 + 2z^{-1} + 2z^{-2} + z^{-3}}{2 + 0.8z^{-1} + 0.49z^{-2} - 0.57z^{-3}}$$

Solⁿ let $H(z) = \frac{B_m(z)}{A_m(z)}$

$$A_m(z) = 2 + 0.8z^{-1} + 0.49z^{-2} - 0.57z^{-3}$$

$$= 2(1 + 0.4z^{-1} + 0.245z^{-2} - 0.285z^{-3})$$

let $A_3(z) = 1 + 0.4z^{-1} + 0.245z^{-2} - 0.285z^{-3}$

$$\alpha_3(0) = 1 \quad \alpha_3(1) = 0.4 \quad \alpha_3(2) = 0.245 \quad \alpha_3(3) = -0.285$$

$$B_3(z) = -0.285 + 0.245z^{-1} + 0.4z^{-2} + z^{-3}$$

we know, $A_{m-1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2}$

for $m=3$

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$A_2(z) = \frac{1 + 0.4z^{-1} + 0.245z^{-2} - 0.285z^{-3} - (-0.285)(-0.285 + 0.245z^{-1} + 0.4z^{-2} + z^{-3})}{1 - (-0.285)^2}$$

$$= \frac{0.9187 + 0.469z^{-1} + 0.359z^{-2} + 0}{0.9187}$$

$$\therefore A_2(z) = 1 + 0.51z^{-1} + 0.39z^{-2}$$

$$\alpha_2(0) = 1 \quad \alpha_2(1) = 0.51 \quad \alpha_2(2) = K_2 = 0.359$$

$$B_2(z) = 0.359 \cdot \cancel{0.359 z^0} + 0.51 z^{-1} + z^{-2}$$

$$\text{For } m=2 \quad A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$\begin{aligned} A_1(z) &= \frac{1 + 0.51 z^{-1} + 0.359 z^{-2} - 0.359 (0.359 + 0.51 z^{-1} + z^{-2})}{1 - (0.359)^2} \\ &= \frac{0.871 + 0.3869 z^{-1} + 0}{0.871} \end{aligned}$$

$$\therefore A_1(z) = 1 + 0.444 z^{-1}$$

$$\alpha_1(0) = 1 \quad \alpha_1(1) = 0.444 = K_1$$

for ladder coeff. let $B_3(z) = 1 + 2z^{-1} + 2z^{-2} + z^{-3}$, $M=3$
on solving we get $b_0=1 \quad b_1=2 \quad b_2=2 \quad b_3=1=c_3$

$$\cancel{c_0 = 1.6665} \quad \cancel{c_1 = 1.6665} \quad \cancel{c_2 = 1.6665} \quad \cancel{c_3 = 1.6665}$$

we know,

$$c_m = b_m - \sum_{i=m+1}^{M=3} c_i \alpha_i(i-m)$$

for $m=2$

$$c_2 = b_2 - \sum_{i=3}^3 c_i \alpha_i(i-2)$$

$$= b_2 - c_3 \alpha_3(1)$$

$$= 2 - 1 \times 0.4$$

$$\therefore c_2 = 1.6$$

for $m=1$
$$C_1 = b_1 - \sum_{i=0}^3 C_i \alpha_i(i-1)$$

$$= b_1 - C_2 \alpha_2(1) - C_3 \alpha_3(2)$$

$$= 2 - 1.6 \times 0.51 - 1 \times 0.245$$

$$\therefore C_1 = 0.939$$

for $m=0$

$$C_0 = b_0 - \sum_{i=1}^3 C_i \alpha_i(i-0)$$

$$= b_0 - C_1 \alpha_1(1) - C_2 \alpha_2(2) - C_3 \alpha_3(3)$$

$$= 1 - 0.939 \times 0.444 - 1.6 \times 0.359 - 1 \times (-0.285)$$

$$\therefore C_0 = 0.293$$

$\therefore$ ladder coeff. are $C_0 = 0.293$ $C_1 = 0.939$ $C_2 = 1.6$ $C_3 = 1$

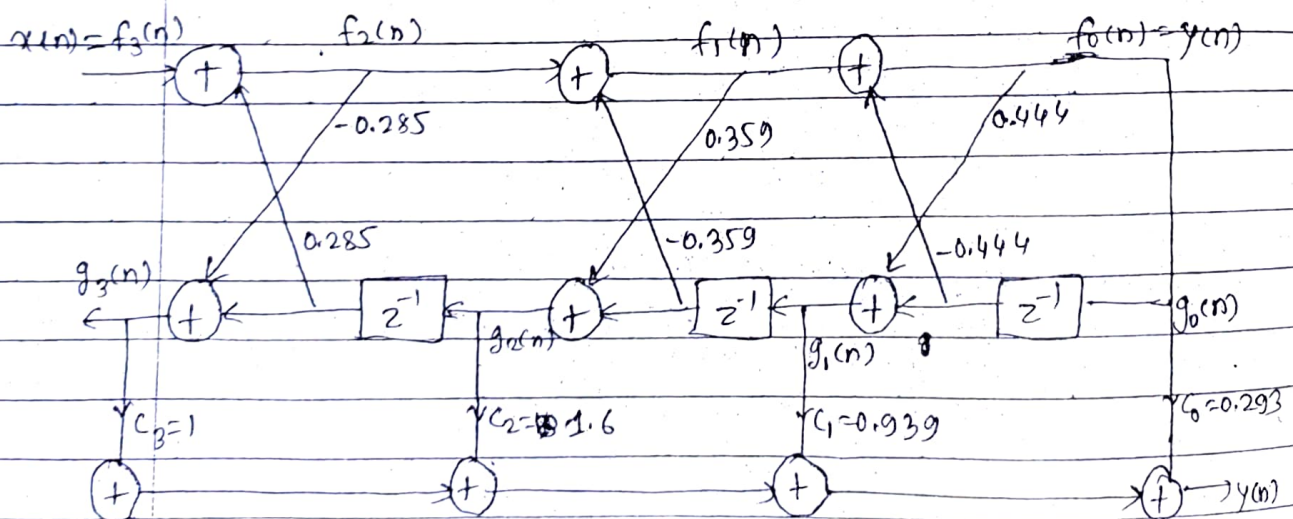


Fig: lattice-ladder structure of pole-zero system

Q. Draw the lattice ladder diagram for the following system
 $H(z) = \frac{2 - 0.6z^{-1} - 0.4z^{-2}}{3 + 0.9z^{-1} + 1.8z^{-2}}$ Is this system stable?

Solⁿ Let $H(z) = \frac{2 - 0.6z^{-1} - 0.4z^{-2}}{3(1 + 0.3z^{-1} + 0.6z^{-2})}$; let $H(z) = \frac{B_m(z)}{A_m(z)}$

Also, let $A_2(z) = 1 + 0.3z^{-1} + 0.6z^{-2}$ where, $A_m(z) = 3(1 + 0.3z^{-1} + 0.6z^{-2})$

$\alpha_2(0) = 1$ $\alpha_2(1) = 0.3$ $\alpha_2(2) = 0.6 = K_2$

$B_2(z) = 0.6 + 0.3z^{-1} + z^{-2}$

we know, $A_{m-1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2}$

for $m=2$

$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2} = \frac{1 + 0.3z^{-1} + 0.6z^{-2} - 0.6(0.6 + 0.3z^{-1} + z^{-2})}{1 - (0.6)^2}$

$\therefore A_1(z) = \frac{0.64 + 0.12z^{-1}}{0.64} \Rightarrow A_1(z) = 1 + 0.1875z^{-1}$
 $\alpha_{1,0} = 1$ $\alpha_{1,1} = K_1 = 0.1875$

$\therefore$ lattice coeff. are, $K_1 = 0.1875$ & $K_2 = 0.6$

for ladder coeff

Let $B_2(z) = 2 - 0.6z^{-1} - 0.4z^{-2}$, $B = 2$

$b_0 = 2$ $b_1 = -0.6$ $b_2 = -0.4 = C_2$

we know, $C_m = b_m - \sum_{i=m+1}^M C_i \alpha_i(i-m)$

for $m=1$, $C_1 = b_1 - \sum_{i=2}^2 C_i \alpha_i(i)$

$= b_1 - C_2 \alpha_2(2)$

$= -0.6 - (-0.4) \times 0.6$

$\therefore C_1 = -0.36$

for $m=0$, $c_0 = b_0 - \sum_{i=1}^2 c_i \alpha_i(i)$

$$= b_0 - c_1 \alpha_1(1) - c_2 \alpha_2(2)$$

$$= 2 - 0.1875 \times (-0.36) - 0.4 \times 0.6$$

$$= 1.8275$$

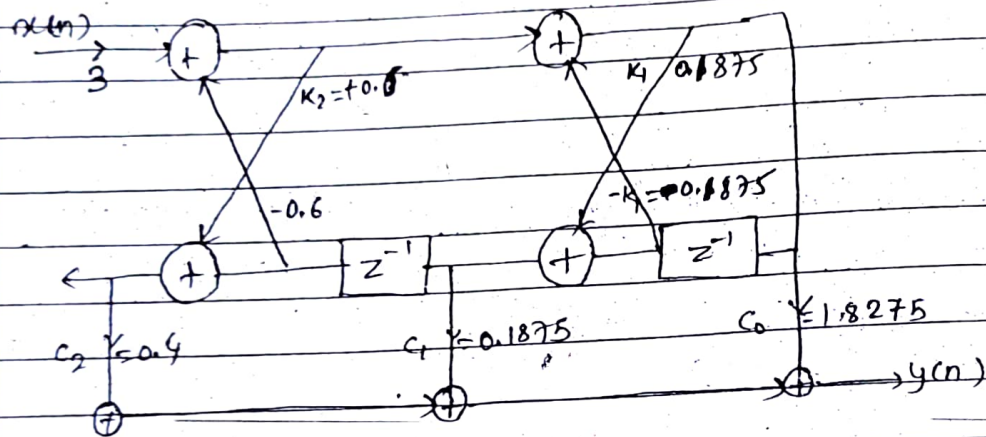


fig: lattice-ladder structure of pole-zero system